

chairman K. Radhakrishnan for whom this Vishu, the beginning of the Malayalee's New Year, is a bad start. The very launch of the Indian-made cryogenic engine is the fulfilment of the dream of the founding father of the Indian space programme, Vikram Sarabhai. He, as the ISRO chief, had initiated the cryo project a little before his death in 1971 but for some reasons the programme died with him. India was forced to look within, in 1993, when Russia joined the Missile Technology Control Regime (MTCR)—a voluntary association of nations which share the goals of non-proliferation of unmanned systems capable of delivering weapons of mass destruction.

The indigenous effort began after five launchers were designed and developed using Russian engines—each costing between \$3 and \$5 million. India had negotiated with Russia since 1988 for the cryogenic engines, where they were developed in the '70s, and finally contracted with Russia in April 1992 when the US struck. Following a compromise formula, Russia agreed to supply seven engines without the transfer of critical technology. The first engine landed in India in 1996 for the planned inaugural GSLV mission in late 1997 or early 1998. At the end of 1993, with Russia joining the MTCR, the terms of the contract were revised.

The deal was to be executed on the Russian side by the Salyut Design Bureau of the Khrunichev State Research and Production Space Centre, Moscow. Salyut opened an office in Chennai because the assembly, autonomous systems tests and comprehensive tests of its engines demanded the presence of at least 50 Russian scientists. In 1998, more than 150 Russians were in India to help

Plans Go Awry

ISRO's next major event is the PSLV-C15 launch in May

— The Polar Satellite Launch Vehicle (PSLV) is ISRO's first operational launch vehicle. It can launch 1,600-kg satellites in 620-km sun-synchronous polar orbit and a 1,050-kg satellite in geosynchronous transfer orbit.

— There have been 15 continuously successful flights of PSLV till September 2009. The Chandrayaan mission saw a PSLV variant carrying an enhanced payload of 1,750 kg.



push the launch programme. GSLV-D1, after initial hiccups, blasted off on April 18, 2001, with a Russian cryo engine.

LPSC project leaders like V. Gnana Gandhi had spent several months at

the Russian facility in the Urals trying to garner information on the cryo engines. "We were more than 30 scientists and had to pick up as much as we could," he says. From one mountain in Russia it was another mountain in southern India where the first big challenge, as he says, was the production and storage of liquid hydrogen.

Backed by a Rs 300-crore grant from the Centre, teams of scientists slogged for several months from the mid-'90s to create the fuel. The space scientists roped in academicians from IIT-Chennai, IISc-Bangalore and IIT-Kharagpur, apart from industry players like Godrej and HAL to work on the fabrication and assembling of key testing systems. The biggest challenge was to store fuels like liquid hydrogen at -250° Celsius (six times colder than Drass in Ladakh where it is -40° Celsius). Working on a cryogenic stage has its risks: handling, storing and pumping the cryogenic fluids at very low temperatures calls for highly advanced technology because they are extremely volatile. If they are stored in tanks or pumped through pipes made of ordinary metals, the metals will become brittle and lubricants will solidify. So LPSC scientists developed new alloys to store liquid hydrogen and liquid oxygen and pump them through pipes. The pipes were made of different metals and had to be welded together.

Aeronautical engineers at HAL, Bangalore, worked on preventing hydrogen leaks. Testing the cryogenic stage on the ground is more difficult than in flight because of the atmospheric pressure on the Earth. ISRO worked with the industry to fabricate tanks weighing less than the Russian ones and made of high-strength aluminium alloy to store cryogenic propellants. Development of booster high-speed turbo-pumps for liquid hydrogen was another big challenge. As hydrogen has low density, the turbo-pumps have to operate at very high speeds—40,000 revolutions a minute, or 700 times every second. At such high speed, proper cooling techniques were developed to keep the bearings cool. As hydrogen leaks eas-

“WE HAVE WORKED HARD FOR the past eighteen years and will soon be back with the next GSLV mission in less than a year.”

K. RADHAKRISHNAN
CHAIRMAN, ISRO

